

Data Scientist with a strong focus on machine learning modeling, evaluation, and error analysis, and hands-on experience designing end-to-end data-driven systems. Experienced in benchmarking models, analyzing generalization behavior, and translating experimental findings into scalable, production-ready solutions.

SKILLS

Programming & Querying	Python, SQL, C/C++, Bash
Data Analysis & ML	NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, XGBoost
Modeling & Evaluation	Feature Engineering, Model Benchmarking, Error Analysis, Evaluation Metrics (Precision, Recall, IoU, Distributional Metrics)
Data Systems & Experimentation	Databricks, MLflow, Jupyter, Git/GitHub, Linux

EDUCATION

Master of Science in Applied Computing (CGPA 4.0/4), University of Toronto Artificial Intelligence Focus	September 2024 - December 2025 Toronto, ON, Canada
Honors Bachelor of Science with High Distinction (CGPA 3.94/4), University of Toronto Specialist in Computer Science, Major in Mathematics, Minor in Statistics	September 2018 - April 2024 Toronto, ON, Canada

TECHNICAL EXPERIENCE

Data Science Intern PyTorch, OpenCV, XGBoost, Databricks, Docker, NVIDIA Triton Magna New Mobility	May 2025 – December 2025 Toronto, ON, Canada
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- Constructed a **patch-based analysis pipeline** for ultra-high-resolution industrial images, transforming raw inputs into window-level predictions suitable for **scalable model evaluation** under compute constraints.
- Achieved measurable improvements in small-defect detection (**+0.19 IoU, +0.30 Precision, +0.22 Recall**) through **data-driven experimentation**, including targeted augmentation strategies, architectural comparison, and systematic stitching optimization.
- Developed a **hybrid modeling workflow** for leather defect analysis (**binary segmentation** → **defect crop extraction** → **XGBoost multi-class severity classification**), enabling **interpretable severity modeling** and structured error analysis across defect types.
- Designed **end-to-end data and model pipelines** using **Databricks, Docker, and NVIDIA Triton** to support reproducible experimentation, scalable inference, and deployment-aware evaluation in production-like settings.

Software Engineer Intern C++, Python, ROS2 Foxy, OpenCV, Linux, Git Magna Electronics	May 2022 - June 2023 Brampton, ON, Canada
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- Improved data quality for vision-based sensing by performing **intrinsic and extrinsic camera calibration**, ensuring accurate sensor alignment for downstream perception analysis and model evaluation.
- Increased robustness of data-processing and perception pipelines by debugging and extending **C++/Python modules**, reducing failure cases under real-world operating conditions.
- Supported production-facing analytics and model validation by managing **version-controlled customer deployments**, and separately contributing to **ROS2 Foxy platform integration on Linux**, maintaining reproducibility and code quality through testing and review workflows.

PROJECTS

INTR: Interpretable Transformer for Image Classification Python, PyTorch, git	September 2024 - December 2024
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- Conducted a **large-scale evaluation study** of INTR across **14 fine-grained datasets** (8 original + 6 extended), combining **accuracy benchmarking** with attention-based diagnostics.
- Analyzed **generalization behavior** by benchmarking INTR against **ResNet-50**, identifying dataset-dependent performance shifts, including up to **+9.2% accuracy gain** on Fish and up to **-12% degradation** on CUB under complex background conditions.
- Performed **interpretability-driven analysis** using **class-specific cross-attention patterns** to attribute model predictions to discriminative regions, supporting error analysis in fine-grained classification.

Artwork Restoration with Generative Image Inpainting Python, PyTorch, git	September 2024 - December 2024
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- Designed a **controlled benchmarking study** to evaluate **3 generative inpainting models** (CO-MOD-GAN, ICT, LaMa) on the **WikiArt dataset**, training on **20k images** and testing on a **5k-image holdout set** with **5 degradation mask types**.
- Quantified restoration performance using **MSE, MS-SSIM, and FID**, revealing clear model trade-offs: LaMa achieved the best **pixel-level and structural fidelity** (MSE 0.01002, MS-SSIM 0.79205), while Co-Mod-GAN produced the most realistic global distributions (**FID 112.155**).
- Conducted **qualitative error analysis** across diverse art styles to identify **systematic failure modes**, including facial distortion, fine-detail degradation, and performance collapse under full occlusion.